Research Question:

- *Must include IV, DV, Units, Measurement tool/Method with uncertainty, and population/system.*
- *Example: "How does the speed of a softball (km/h), measured using a Bushnell 101911 Velocity Speed Gun (±2 km/h), differ given the type of throw used (underarm, sidearm and overarm) by U19 varsity softball players."*
- *Higher-band: Specific, measurable, reproducible RQs.*

Background Information (750 Words):

- *Provide a scientific rationale linking IV and DV.*
    - *DO NOT spend words defining the IV and DV, use peer review articles that align with this.*
- *Focus on peer-reviewed literature that directly connects to your RQ.*
    - *Explains why there might be a link between the independent and dependent variables that is worth investigating.*
- *Use at least four peer-reviewed sources here.*
- *Higher-band: Shows how prior studies underpin the chosen design, not just definitions.*

Hypothesis:

- Alternate Hypothesis:*Predict expected outcome with justification from literature (120 Words Max).*
- Null Hypothesis: There will be no statistically significant effect of the *{independent variable}* on the *{dependent variable}*.

Variables:

Independent Variable:

- *Fully describe these with context and if possible/practical an image.*
- *The variable that is being changed, simply put what is being tested against what. For example: three different types of shoes (Anova*

statistical test), Difference in male/female sprints (T-Test), Hamstring length against sprint time (correlation).
Here is an example:

Independent variable: The type of footwear worn by the subjects. Subjects will be tested under three conditions:

- *Wearing a pair of trainers normally suited for running.*
- *Wearing a pair of soccer shoes designed for use on turf.*
- *Wearing a pair of soccer shoes designed for use on hard ground.*

Note: that subjects were asked to provide their own footwear. All footwear was verified to ensure it met the characteristics of the condition. Samples of the footwear shown below:

Figure 1 - Typical sole pattern of shoes in each of three conditions of the independent variable. From left to right - trainers, turf shoes, hard ground cleats.



Dependent Variable:

- *Fully Describe the process of measurement (not just what you are measuring with).*
- *Photographic evidence.*
- *Uncertainties of measuring devices, make sure these are factual.*
- *Units measured in.*
- *Include references to literature if you have used it to inform your study, especially if you are using a standardised test.*

Controlled Variables:

A description of the controlled variables, and explanation of how they were controlled and how a lack of control could have impacted the results.

If you can reference how or why it is controlled through research done into other studies this is higher band.

Must include traditional controls (time of day, warm-up, environment) and bias controls:

- *Order effects (use randomisation or counterbalancing).*
- *Practice effects (pilot tests or familiarisation trials).*
- *Placebo effects (e.g., blinding participants when testing).*
- *Observer bias (objective tools such as timing gates instead of stopwatches).*

These must be tabled.

Controlled Variable	How is it Controlled:	Why it is Controlled

Confounding Variables:

A description of the confounding variables, explaining why they could not reasonably be controlled, and how they may confound the results.

Uncontrolled Variable	How could it affect experimentation?	Steps taken to mitigate that effect?

Safety and Ethics:

Physical Safety: (You Might choose to do a table for physical)

Hazard	Hazard Rating (1-5)	Who might be harmed and how?	What are you already doing to control the risks?

Physical:

- State that a PAR-Q has been done, a blank copy should go in the Appendix.
- State that a Student Consent form has been filled in, a blank copy should go in the appendix.
- The testing protocol should have been checked by the adviser/teacher.
- The testing protocol should adhere to the IB Experimental Science Guidelines.
- 1st Aid should be on site for the duration of testing e.g. the school nurse.
- The area should be checked and clear of any physical dangers.
- Participants should be checked to be free from injury and capable of the testing protocol.

Ethical Safety:

- Participants have a right to anonymity, all names and personal identifiers have been removed from the writing.
- Participants have the right to withdraw from the experiment and withdraw the data collected that adheres to them at any time.
- Participants will be briefed in accordance with the experimentation protocol.
- Participants will be briefed, debriefed, warmed up and cooled down at each stage of testing protocol.

Environmental Safety:

- The majority of the time there is no environmental impact or safety. If there is none, students should say that there is none. This shows that they have at least considered it.
- Some factors students might consider could be, not using paper questionnaires, testing outside instead of in a gymnasium (Electricity and Aircon).

Higher-band: Explicit statements of safety and ethics in the report, nothing assumed.

Methodology:

Rationale:

Why are we doing this testing and what has made us come to the conclusion of doing this testing?

What has made you choose these particular variables that you have (namely IV & DV).

The population you have chosen is an indicator or indicative of what based upon the study's purpose.

Selection Criteria:

What method was used to select your participants?

Why was that method chosen, justification?

How did that help with your experimentation by making those choices?

Materials

These only have to be listed if they are measuring tools (as you need to list uncertainty) or Non-generic Items, for instance a EKG Sensor.

Justification of Uncertainty:

Why do you have uncertainty and what that justifies for the testing?

Have you got that justification from somewhere e.g. peer review article, or have you made an educated guess?

If you guess, how did you come to that conclusion?

Pre-Test Protocol:

Your initial testing protocol

How did this work and what are you changing if anything for the actual protocol?

How did this inform what you are going to do?

Test Set-up Procedure: (A picture is always good).

A picture of the set up is always worthwhile and helps to give context to the experimentation.

If you are using a basic testing protocol, e.g the wall ball toss for coordination, this only has to be briefly written, but should be referenced.

If your testing protocol is novel, then a full description should be here, this will save words in the testing protocol section.

Testing Protocol:

How was the testing completed, step by step.

This needs to be bullet pointed with enough description to be replicated.

All methodologies should start with a briefing and warm-up (these can be put into the appendix).

All methodologies should finish with a cool-down and debrief, (again can be in appendix).

Data Analysis:

Raw Data:

Present clearly in tables, either in the body (if small) or appendix (if large).

If the table is too big, provide a screenshot sample (this would only be for data tables that are 100's of data points).

If you used a computer program such as Vernier Graphical Analysis, provide a couple of screen shots as well as the raw data. Put full screen shots in the appendix.

Include all raw measurements with units and uncertainties.

Use consistent decimal places, two decimal places for most experimentation.

If you have relevant Qualitative data, write this in bullet points.

Commentary on Raw Data:

Identify anomalies (injuries, distractions, equipment errors).

Note participant withdrawals or missing data, and explain why.

Comment on bias effects if relevant (fatigue, motivation, practice).

Method Of processing:

The method of testing and the rationale for that testing should be explained and justified.

Analysed Data Tables:

Present means, SD, sample size, units, and uncertainties.

Label tables clearly with descriptive headings.

Interpretation of Table:

Explain what patterns or trends the data shows.

Identify key comparisons (e.g., "Group A had faster average sprint times than Group B").

Mention any anomalies or unexpected results.

Coefficient of Variation:

Work out the "coefficient of variation."

Interpretation of variation:

What does that mean?

Graphs:

Choose graph type appropriate to data (scatter, line, bar).

Always include error bars (SD or SE).

Axes labelled with variables, units, uncertainties.

Provide clear descriptive title.

Interpretation of Graph:

Describe the overall pattern or trend.

Compare groups or conditions.

Link trends back to the research question.

Comment on whether the trend suggests support or contradiction of the hypothesis.

Statistical Test:

Name the test used and justify why it was appropriate.

Show or describe test result (p-value, correlation coefficient, F-ratio, etc.).

Interpretation of Test:

Explain what the result means in context (e.g., "The difference was statistically significant, therefore null hypothesis rejected").

Relate back to the graph and table interpretations.

Higher-band Expectations:

Each table, graph, and statistical test includes its own interpretation, not left to the conclusion.

Interpretations explain what the data shows and what it means in relation to the RQ.

Comments on measurement tool uncertainty, outliers, and possible participant bias are integrated into the discussion of results.

Conclusion:

Direct Answer to the Research Question:

- State clearly what the data shows in relation to the RQ.
- Use processed data (means, SD, statistical results) to justify the conclusion.
- Build directly on interpretations already made from tables, graphs, and statistical tests.
- Discuss the strength of the conclusion.
- Recognise contextual limitations

Hypothesis Outcome:

- Explicitly state whether the null hypothesis is accepted or rejected.
- Support with evidence from inferential statistics (e.g., "$p < 0.05$, therefore null rejected").

Comparison with Literature:

- Bring in **at least two new peer-reviewed** studies not used in the background information.
- Highlight agreement and disagreement with your findings.
- Discuss reasons for differences (population, measurement tools, context).

Practical Implications:

Apply findings to real-world SEHS practice:

Training design.

Injury prevention.

Coaching strategies.

Health and rehabilitation.

Be specific — identify who benefits and how.

Higher-band Expectations:

Clear, confident answer to the RQ.

Direct use of data and stats, not vague statements.

Literature integrated to show how findings fit into scientific context.

Consideration of generalisability: e.g., "Findings apply to adolescent male athletes but may not transfer to adults or female athletes."

Practical implications are meaningful and realistic, not generic.

Evaluation:

Limitations of Method and Data:

- Discuss weaknesses in design, data collection, and measurement tools.
- Mention representativeness and reproducibility.

Error Types:

Distinguish between:

- Systematic errors (consistent measurement error, e.g., stopwatch reaction time).
- Random errors (inconsistencies from trial to trial).
- Explain how each influenced results and conclusions.
- Uncontrolled Variables and Biases
- Identify confounding variables that could not be fully controlled.

If present explicitly address:

- Order effects (fatigue, learning).
- Practice effects (improvement through repetition).
- Placebo effects (participants changing effort if they believe a treatment is effective).

- Observer bias (researcher influence, subjective measurement).
- Explain their likely influence on results.

Improvements:

- Provide specific, realistic improvements such as:
- Randomising or counterbalancing trial order.
- Including familiarisation trials.
- Using blinding for participants/assessors where possible.
- Using objective measures (e.g., electronic timing gates vs. stopwatch).
- Increasing sample size only if justified (e.g., improves statistical power).
- Link improvements back to identified weaknesses.

Reflection on Validity and Reliability:

- Discuss whether the investigation produced valid results that answer the RQ.
- Consider whether results could be replicated reliably by others.
- Comment on whether findings can be generalised beyond the sample.

Higher-band Expectations:

- Evaluation is detailed and specific to the study, not generic.
- Systematic and random errors are clearly separated.
- Biases are named, explained, and improvements suggested.
- Improvements are justified and sometimes supported with literature.
- Reflection shows depth: validity, reliability, reproducibility, and scope of findings.

Use this table to help you:

Example:

Limitation	***Effect on results***	***Validity / Reliability***	***Improvement***	***Why it works***
Small sample size	*Increased variability*	*Reliability ↓*	*Increase participants*	*Reduces random error*